

بسم الله الرحمن الرحيم

فاطمه اثباتی

گزارش چهارم – مهندسی داده

دی ماه ۱۴۰۳

پیکربندی trino برای اتصال به hive

```
sudo mkdir /etc/trino/catalog
```

```
sudo nano /etc/trino/catalog/hive.properties
```

```
connector.name=hive
```

```
hive.metastore.uri=thrift://<metastore-host>:<port>
```

```
nano /usr/local/hive/conf/hive-site.xml
```

```
<property>
```

```
  <name>hive.metastore.uris</name>
```

```
  <value>thrift://<metastore-host>:<port></value>
```

```
</property>
```

```
sudo chown -R trino:trino /etc/trino/catalog
```

```
sudo chmod -R 755 /etc/trino/catalog
```

```
wget --no-check-certificate https://search.maven.org/remotecontent?filepath=io/trino/trino-cli/436/trino-cli-436-executable.jar -O trino-cli-436-executable.jar
```

```
sudo mv trino-cli-436-executable.jar /usr/local/bin/trino
```

```
sudo chmod +x /usr/local/bin/trino
```

```
trino --version
```

```
[esbati@vbox ~]$ sudo chmod +x /usr/local/bin/trino
[esbati@vbox ~]$ trino --version
Trino CLI 436
[esbati@vbox ~]$
```

trino --server localhost:8080 --catalog hive --schema default

```
[esbati@vbox ~]$ trino --server localhost:8080 --catalog hive --schema default
trino:default> SHOW CATALOGS;
Catalog
-----
hive
system
(2 rows)

Query 20250105_141542_00005_b3pk5, FINISHED, 1 node
Splits: 7 total, 7 done (100.00%)
0.37 [0 rows, 0B] [0 rows/s, 0B/s]
trino:default>
```

چالش:
مشکل زدن query

```
trino:default>
trino:default> CREATE TABLE test_table (
->     id INT,
->     name VARCHAR,
->     created_at TIMESTAMP
-> );
->
Query 20250105_141747_00009_b3pk5 failed: MaxMessageSize reached

trino:default> INSERT INTO test_table VALUES
-> (1, 'Alice', CURRENT_TIMESTAMP),
-> (2, 'Bob', CURRENT_TIMESTAMP),
-> (3, 'Charlie', CURRENT_TIMESTAMP);
->
Query 20250105_141802_00010_b3pk5 failed: MaxMessageSize reached
```

sudo nano /etc/trino/config.properties

```
# Maximum memory for queries
query.max-memory=4GB
query.max-memory-per-node=1GB
```

sudo service trino restart

```
sudo nano /etc/trino/catalog/hive.properties
hive.metastore.thrift.client.max-message-size=104857600
sudo service trino restart
```

cat /var/log/trino/server.log

none of them can not solve the problem so I search that to find out the answer in github

```
Logging initialized using configuration in jar:file:/usr/local/hive/lib/hive-common-3.1.3.jar!/hive-log4j2.properties Async: true
Exception in thread "main" java.lang.RuntimeException: org.apache.hadoop.hdfs.server.namenode.SafeModeException: Cannot create directory /tmp/hive/esbati/4009dfad-80fc-45d4-ba7a-c8329ee1c54e. Name node is in safe mode.
Resources are low on NN. Please add or free up more resources then turn off safe mode manually. NOTE: If you turn off safe mode before adding resources, the NN will immediately return to safe mode. Use "hdfs dfsadmin -safemode leave" to turn safe mode off. NamenodeHostName:localhost
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.newSafeModeException(FSNamesystem.java:1612)
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.checkNameNodeSafeMode(FSNamesystem.java:1599)
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.mkdirs(FSNamesystem.java:3437)
at org.apache.hadoop.hdfs.server.NamenodeRpcServer.mkdirs(NamenodeRpcServer.java:1166)
at org.apache.hadoop.hdfs.protocolPB.ClientNamenodeProtocolServerSideTranslatorPB.mkdirs(ClientNamenodeProtocolServerSideTranslatorPB.java:742)
at org.apache.hadoop.hdfs.protocol.proto.ClientNamenodeProtocolProtos$ClientNamenodeProtocol$2.callBlockingMethod(ClientNamenodeProtocolProtos.java)
at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:621)
at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:589)
at org.apache.hadoop.ipc.ProtobufRpcEngine2$Server$ProtoBufRpcInvoker.call(ProtobufRpcEngine2.java:573)
at org.apache.hadoop.ipc.RPC$Server.call(RPC.java:1213)
at org.apache.hadoop.ipc.Server$RpcCall.run(Server.java:1089)
at org.apache.hadoop.ipc.Server$RpcCall.run(Server.java:1012)
at java.security.AccessController.doPrivileged(Native Method)
at javax.security.auth.Subject.doAs(Subject.java:422)
at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
at org.apache.hadoop.ipc.Server$Handler.run(Server.java:3026)

at org.apache.hadoop.hive ql.session.SessionState.start(SessionState.java:651)
at org.apache.hadoop.hive ql.session.SessionState.beginStart(SessionState.java:591)
at org.apache.hadoop.hive.cli.CliDriver.run(CliDriver.java:747)
at org.apache.hadoop.hive.cli.CliDriver.main(CliDriver.java:683)
at sun.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.lang.reflect.Method.invoke(Method.java:498)
at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
Caused by: org.apache.hadoop.hdfs.server.namenode.SafeModeException: Cannot create directory /tmp/hive/esbati/4009dfad-80fc-45d4-ba7a-c8329ee1c54e. Name node is in safe mode.
Resources are low on NN. Please add or free up more resources then turn off safe mode manually. NOTE: If you turn off safe mode before adding resources, the NN will immediately return to safe mode. Use "hdfs dfsadmin -safemode leave" to turn safe mode off. NamenodeHostName:localhost
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.newSafeModeException(FSNamesystem.java:1612)
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.checkNameNodeSafeMode(FSNamesystem.java:1599)
at org.apache.hadoop.hdfs.server.namenode.FSNamesystem.mkdirs(FSNamesystem.java:3437)
```

sudo nano /etc/hive/conf/hive-site.xml

```
<property>

  <name>hive.metastore.thrift.max.message.size</name>

  <value>104857600</value> <!-- 100 MB -->

</property>

<property>

  <name>hive.server2.thrift.max.message.size</name>

  <value>104857600</value> <!-- 100 MB -->

</property>
```

pskill -f hive

hive --service hiveserver2 &

Exit Safe Mode

hdfs dfsadmin -safemode leave

hdfs dfsadmin -safemode get

Run the following command to take the NameNode out of safe mode

But nothing change so I think the problem is because of my system lack of storage

```
[esbati@vbox tools]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        5.3G   0  5.3G   0% /dev
tmpfs           5.3G   0  5.3G   0% /dev/shm
tmpfs           5.3G  8.7M  5.3G   1% /run
tmpfs           5.3G   0  5.3G   0% /sys/fs/cgroup
/dev/sda2       10G   8.6G  1.5G  86% /
/dev/sda5       8.0G   4.4G  3.7G  55% /home
/dev/sda1      1014M  258M  757M  26% /boot
tmpfs          1.1G   0  1.1G   0% /run/user/1000
[esbati@vbox tools]$
```

چالش ssh و دیدن صفحه وب در سیستم میزبان:

Guest

`sudo systemctl stop firewalld`

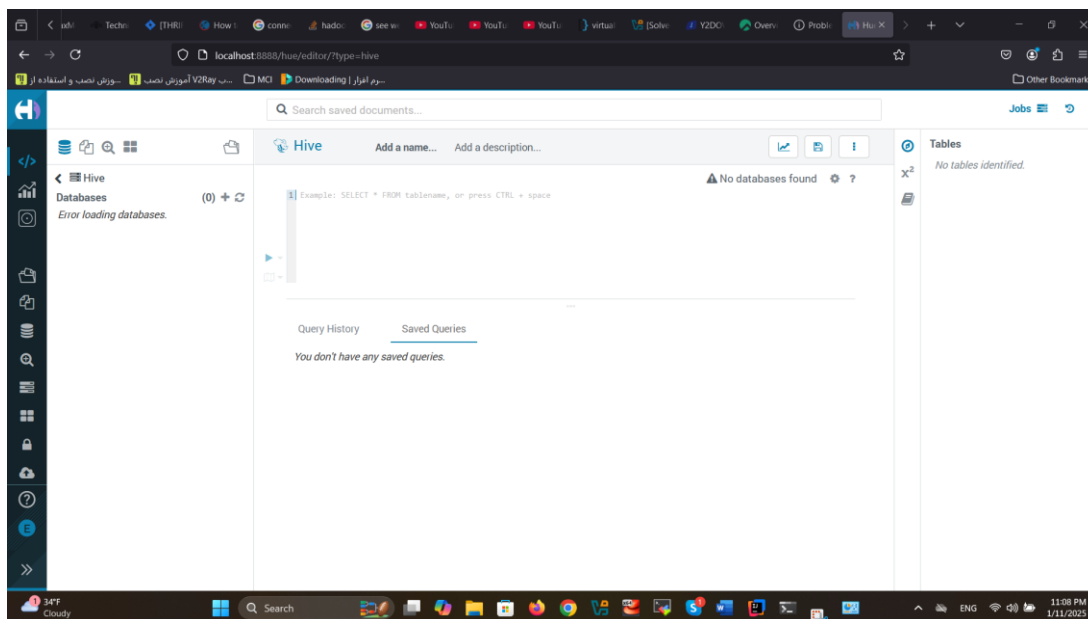
`sudo systemctl disable firewalld`

host

stop firwall from setting for windows

And I also did this when ssh to guest machin via cmd

Activate the web console with: `systemctl enable --now cockpit.socket`



مشکل فایل دیتای tcp ds

<https://github.com/kcheeeung/hive-benchmark>